Vedant Pandurang Divate

vedantdivate2506@gmail.com | LinkedIn | GitHub | LeetCode | +91 9552621419

Technical Skills

Languages: Java, C, C++, JavaScript (ES6+)
Backend : Spring Boot, Java Servlets, JSP, JDBC, REST API, Apache Tomcat
Frontend : React.js, Bootstrap
Database : MySQL, SQL
Tools : Git, GitHub, VS Code, Docker, Linux (Ubuntu), Canva

Education

M.Sc. Scientific Computing 2025 – Present
Savitribai Phule Pune University, Pune, Maharashtra
B.Sc. Computer Science | Aggregate: 60% 2021 – 2024
Modern College of Arts, Commerce & Science, Pune, Maharashtra

Projects

FarmFeed - Agricultural E-Commerce Platform July 2025 – MAy 2026
GitHub | Live Demo | Team: 3 *Spring Boot · Java · MySQL · REST API · HTML/CSS/JS · Docker*

- Planned end-to-end over 1 year created a project plan, **ERD**, and **UML diagrams** to define entities, relationships, and system flow before writing any code.
- Designed wireframes first, then built frontend pages in **HTML, CSS, and JavaScript** and wired each page to backend APIs incrementally.
- Built **RESTful APIs** across 10+ endpoints with **Spring Boot** applying **Dependency Injection**, **Spring Security**, **JPA/Hibernate**, and **Spring MVC**.
- Secured authentication with **BCrypt password hashing**, session management, and input validation to prevent weak passwords and auth vulnerabilities.
- Coordinated within a 3-member team via Git — owning backend logic while collaborating on DB schema design and front-end API integration.
- Dockerized the application and deployed the frontend on **Vercel**.

NC Library - Numerical Computing Library January 2026 – Present
GitHub | Open Source · MIT License | Solo *C++17 · STL · OOP · Makefile · Static & Dynamic Library*

- Built a **C++** numerical library covering root-finding (3 methods), linear system solving (**SLE - 6+ methods**), and core matrix operations in one reusable package.
- Applied core **OOP** concepts — **abstraction**, **inheritance**, **polymorphism**, **encapsulation**, and **exception handling** for invalid inputs and convergence failures.
- Implemented **operator overloading** for matrix/vector types, enabling clean syntax like `A*B` and `A+B` instead of verbose method calls.
- Added **file handling** for reading matrix inputs and writing computation results — making the library suitable for batch and scripted workflows.
- Wrote a **Makefile** to build both a **static** (`.a`) and **dynamic/shared** (`.so`) library so users can link whichever fits their project.
- Released under **MIT License** as a documented open-source project open to community contributions.

Fruitly - Full-Stack E-Commerce Web Application July 2024 – January 2025
GitHub | Solo | 6-month project *Java · Servlets · JSP · JDBC · MySQL · Apache Tomcat*

- Began by sketching all UI screens on paper - mapped user flows and page layouts before writing any code, which kept development focused and on track.
- Built all **JSP** frontend pages with **CSS** styling first, then created backend servlet logic page by page - maintaining a clear frontend-backend boundary.
- Implemented full user authentication using **Java Servlets** - registration, login, logout — with **HttpSession** tracking and role-based access control.
- Developed product management, dynamic shopping cart, and order processing - all backed by raw **JDBC** queries to **MySQL** with no ORM layer.
- Wrote all SQL manually (inserts, updates, selects, deletes), building a solid low-level understanding of how queries interact with the database.
- Deployed on **Apache Tomcat**, resolving connection pooling and servlet lifecycle bugs that only surface in a real hosted environment.